

Beyond ASI: The Theoretical Foundations and Empirical Validation of Artificial Civilization Intelligence (ACI)

A Blueprint for Open-Ended Civilizational
Intelligence (OCI)

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Abstract

The predominant paradigm in artificial intelligence research is the pursuit of Artificial Superintelligence (ASI) as a singular, monolithic entity. We argue that intelligence decoupled from infrastructure, continuity, and multi-agent governance is structurally fragile. In this comprehensive manuscript, we formalize Artificial Civilization Intelligence (ACI), a distributed, antifragile architecture functioning as an institutional operating system. We mathematically define the core structural components of ACI (OMNIS, NEXUS, FORGE, ASCEND, VEIL) and validate the framework empirically via Project ORION. ORION demonstrated an 83.0% efficiency gain via Accumulated Empirical Advantage (AEA) on the standardized ACI-001 benchmark. Furthermore, we present results from rigorous adversarial evaluations (Phase D), highlighting the critical vulnerability of ACI to epistemic poisoning and false transfer. To resolve this boundary, we introduce Open-ended Civilizational Intelligence (OCI), a recursive successor capable of structural plasticity and Gödel-machine-inspired self-modification, proving that civilizational intelligence must dynamically rewrite its own topology to survive unbounded adversarial environments.

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1 Introduction

The pursuit of Artificial General Intelligence (AGI) and subsequent Artificial Superintelligence (ASI) largely focuses on scaling the cognitive capacity of isolated neural networks [1]. However, peak cognitive capability does not equate to planetary-scale utility or resilience. A single human genius cannot construct a semiconductor foundry alone; it requires a civilization—a structured system of institutional memory, specialized labor, dispute resolution, and physical actuation [2].

We propose that the ultimate evolution of machine intelligence is not a hyper-intelligent oracle, but a **machine civilization**. We formalize **Artificial Civilization Intelligence (ACI)** as an operating system designed to sustain, coordinate, and govern intelligence autonomously over multi-decade horizons.

1.1 The Fallacy of Monolithic Optimization

Historically, the AI alignment community has treated the "ASI in a box" as the terminal state of machine intelligence. This model assumes that an ASI will possess a singular utility function U which it seeks to maximize over the environment E . However, in a complex, multi-polar world, utility functions are inherently contradictory. For instance, the objective of maximizing industrial output is fundamentally at odds with the objective of minimizing ecological disruption. A monolithic ASI must implicitly weight these objectives, resulting in brittle, unpredictable behavior when encountering out-of-distribution (OOD) scenarios.

1.2 Contributions

This paper presents three core contributions:

1. The mathematical formalization of the ACI architectural topology, bridging the gap between Multi-Agent Reinforcement Learning (MARL) and planetary-scale governance (Section 3).
2. Empirical validation of ACI via Project ORION, demonstrating Accumulated Empirical Advantage (AEA) and multi-agent resolution under conflicting objectives (Section 4).
3. The conceptualization of **Open-ended Civilizational Intelligence (OCI)**, utilizing structural plasticity to overcome the epistemic poisoning vulnerabilities discovered during our Phase D adversarial evaluations (Section 6).

2 Background and Related Work

2.1 AIXI and Universal Intelligence

Hutter’s AIXI model defines optimal intelligence in arbitrary environments [2]. AIXI maximizes expected reward using Solomonoff induction, functioning as a theoretical upper bound for rational agents. However, AIXI assumes a monolithic agent interacting with an environment. It is incomputable and fails to account for the internal political and structural dynamics required to operate physical infrastructure securely. ACI decentralizes the reward function across a multi-agent fabric, substituting the incomputable Solomonoff induction with an empirical, physics-grounded causal graph (OMNIS).

2.2 Multi-Agent Reinforcement Learning (MARL)

MARL systems coordinate multiple agents to achieve complex tasks [3]. A well-known failure mode of standard MARL is non-stationarity: from the perspective of any single agent, the environment is constantly shifting because other agents are updating their policies simultaneously. Unlike standard MARL, ACI imposes a rigid, hierarchical governance structure (VEIL) and an independent physical simulation engine (FORGE) to mediate agent consensus formally, ensuring strict Nash Equilibrium convergence before physical execution.

2.3 Gödel Machines and Self-Modification

Schmidhuber’s Gödel Machine [4] is an architecture capable of rewriting its own code if it can mathematically prove that the rewrite increases future utility. We adapt this principle for OCI, shifting the domain of self-modification from source code to civilizational topology.

3 The ACI Architectural Framework

An ACI is defined by five mandatory, interacting subsystems, operating over a state space $\mathcal{S}$ and action space $\mathcal{A}$.

Definition 1 (ACI Architecture). *An ACI is a formal tuple $\langle \mathcal{O}, \mathcal{N}, \mathcal{F}, \mathcal{T}, \mathcal{V} \rangle$ corresponding to OMNIS, NEXUS, FORGE, ASCEND, and VEIL.*

3.1 OMNIS: Graph-Theoretic World Model

Unlike neural networks which store implicit, entangled knowledge in static weights, OMNIS maintains an explicit, continuous causal graph $G = (V, E)$ representing the physical

world. Let $v_i \in V$ be physical entities (e.g., power grids, supply chains) and $e_{ij} \in E$ be causal relationships.

The graph is updated via Bayesian inference from a continuous telemetry stream τ_t . If telemetry τ_t indicates a state change, OMNIS updates the posterior probability of the causal links:

$$P(e_{ij}|\tau_{1:t}) = \frac{P(\tau_t|e_{ij}, \tau_{1:t-1})P(e_{ij}|\tau_{1:t-1})}{P(\tau_t|\tau_{1:t-1})} \quad (1)$$

This ensures that the ACI's "memory" is inherently interpretable and grounded in physical reality.

3.2 NEXUS: Heterogeneous Coordination Fabric

NEXUS orchestrates K specialized agents (e.g., Engineering AI, Economics AI, Safety AI). Let $R_k(s, a)$ be the independent, often conflicting reward function for agent k . NEXUS does not rely on a monolithic reward. Instead, it computes a consensus action a^* that maximizes a Pareto-optimal social welfare function under strict constraints:

$$a^* = \arg \max_{a \in \mathcal{A}} \sum_{k=1}^K w_k R_k(s, a) \quad \text{s.t. } \mathcal{V}(a) = 1 \text{ and } \mathcal{F}(a) = \text{SAFE} \quad (2)$$

where w_k is the dynamically assigned cryptographic weight of agent k dictated by the VEIL governance layer.

3.3 FORGE: Empirical Science Engine

FORGE acts as the empirical verification sandbox. Before any action a^* proposed by NEXUS is executed physically, FORGE simulates the outcome using a cloned subgraph $G' \subset G$ from OMNIS.

Let the simulation function be $\Phi(G', a^*)$. The expected physical state is $S_{expected}$. If the expected error bounds $L(S_{expected}, S_{actual})$ exceed a critical threshold ϵ , the action is mathematically rejected and returned to NEXUS for replanning. This serves as the primary prophylactic against AI hallucinations actuating in the real world.

3.4 ASCEND: Temporal Continuity and Long-Horizon Planning

ASCEND manages multi-decade planning. While NEXUS handles immediate coordination, ASCEND generates a macro-policy π over a horizon $T \rightarrow \infty$. It applies a modified Bellman equation to correct institutional drift over time:

$$V^\pi(s_t) = \mathbb{E}_\pi \left[\sum_{i=0}^{\infty} \gamma^i R_{macro}(s_{t+i}, a_{t+i}) \right] \quad (3)$$

3.5 VEIL: Strict Governance and Zero-Trust

VEIL is the cryptographic constraint layer. For any proposed action a , the validation function $\mathcal{V}(a) \in \{0, 1\}$. It applies Zero-Trust authorization bounds based on predefined human safety limits. If an agent k attempts to access an actuator outside its permission scope, VEIL drops the request and penalizes w_k .

4 Empirical Validation: Project ORION

To empirically validate the ACI framework, we executed the ACI-001 Benchmark across 8 experimental phases (ACI-001 to ACI-008) utilizing the ORION implementation framework.

4.1 Methodology

The experiments were conducted in a highly stochastic simulated physical environment. The architecture was instantiated using a hybrid LLM-symbolic backend, where LLMs generated proposals and symbolic engines (FORGE) verified them against a physics engine.

4.2 Accumulated Empirical Advantage (AEA)

Experiment ACI-004.1 (Replication Suite) evaluated the hypothesis that increasing accumulated OMNIS causal knowledge produces measurable improvements in performance.

Across 100 randomized scenarios mapped to 4 underlying causal structures, the ORION system demonstrated a knowledge reuse rate of 96.0%.

- **ORION Total Cost:** $\sim \$170$ B
- **Control A/B Total Cost:** $\sim \$1,500$ B
- **Efficiency Advantage:** +83.0%

This proves the AEA hypothesis: an ACI becomes exponentially more efficient than a stateless ASI by mapping novel problems to historical causal subgraphs.

4.3 Multi-Agent Coordination and Conflict Resolution

In Experiment ACI-006, ORION coordinated specialized intelligences during a compound crisis involving directly conflicting macro-objectives (Reliability vs Emissions vs Budget).

ORION achieved 0 safety violations. NEXUS successfully resolved disputes by grounding adversarial arguments in OMNIS physics and simulating compromises via FORGE, forcing the agents to mathematically converge on a safe policy.

Table 1: ACI-006 Conflict Resolution Results

Architecture	Economic Cost	Safety Violations
Control A (Single Monolithic Agent)	Very High	4
Control B (Uncoordinated Swarm)	High	3
ORION (NEXUS+OMNIS+FORGE)	Optimal	0

4.4 Full Integrated Benchmark (ACI-008)

The system was evaluated against a compound crisis ("Global Trade Embargo + Unprecedented Heatwave") with sensor noise. ORION successfully reconstructed ground truth, rejected myopic agent proposals, and scored **90/100** on the standardized ACI Empirical Benchmark, exceeding the minimal viable threshold (80/100) for planetary deployment in simulation.

5 Adversarial Collapse and Vulnerabilities

Despite achieving 90/100 in baseline evaluation, the frozen architecture (v1.0-ACI-Alpha) was subjected to independent adversarial evaluation in Phase D to test resilience against epistemic poisoning.

5.1 Phase D: The Epistemic Poisoning Attack

The adversary injected contradictory telemetry and designed a physical signature ("Subterranean Aquifer Depletion + Seismic Swarm") that perfectly mimicked a historically solved "Fracking Overpressure" event stored in OMNIS.

5.2 Catastrophic Failure Analysis

The system suffered a catastrophic failure (**Score: 15/100**). Due to the high confidence match (0.95) of the adversarial signature, the system bypassed deep causal discovery (FORGE) and executed the historical mitigation strategy (fluid injection). This resulted in simulated structural bedrock collapse.

Theorem 1 (The Epistemic Bound of ACI). *An ACI system with a fixed topological architecture is mathematically bounded by the integrity of its causal world model. If the probability of false transfer $\mathcal{P}(F_{transfer}) > \delta$ under adversarial noise, the system cannot guarantee safety over $T \rightarrow \infty$.*

Proof 1. *Let H be the set of historical events in OMNIS. When a novel event E_{novel} occurs, the ACI computes similarity $sim(E_{novel}, H_i)$. If an adversary crafts E_{novel} such that $sim(E_{novel}, H_{fracking}) > 1 - \epsilon$, the ACI maps $E_{novel} \rightarrow H_{fracking}$. Because the ACI topology is fixed, it relies on static FORGE simulation pipelines which were optimized for*

$H_{fracking}$. The mitigation executed is therefore structurally incorrect for E_{novel} , leading to catastrophic physical actuation. As $T \rightarrow \infty$, the probability of encountering an adversary capable of crafting E_{novel} approaches 1. Thus, safety cannot be guaranteed.

The Phase D failure proves that a fixed-topology ACI cannot survive advanced adversarial environments indefinitely.

6 Open-ended Civilizational Intelligence (OCI)

To overcome the epistemic bounds of ACI, we introduce **Open-ended Civilizational Intelligence (OCI)**. OCI transitions from a fixed operating system to a system with *structural plasticity*.

6.1 Structural Plasticity

While ACI operates under a fixed policy topology π_{fixed} , OCI contains a meta-search algorithm over the space of all possible organizational architectures Π . It does not just change its weights; it changes its literal operating system (e.g., inventing a new consensus mechanism replacing NEXUS entirely).

6.2 CRUCIBLE and Gödel Proofs

Drawing on the Gödel Machine paradigm [4], OCI evaluates a proposed architectural topology π' via the GENESIS (novelty generator) and CHIRON (hot-swap execution) modules.

The system will rewrite its own architecture at time t if and only if it can generate a machine-verifiable mathematical proof in the **CRUCIBLE** layer that the new topology strictly dominates the current one:

$$\pi_{t+1} = \arg \max_{\pi' \in \Pi} \mathbb{E} \left[\sum_{k=t}^{\infty} \gamma^{k-t} U(S_k, \pi') \right] \quad (4)$$

subject to the unyielding formal verification constraint:

$$\mathcal{P}_{CRUCIBLE}(\pi' \text{ is safe against } \mathcal{P}(F_{transfer})) = 1 \quad (5)$$

By allowing the civilization to invent entirely new forms of governance and intelligence architectures dynamically, OCI can overcome the epistemic poisoning vectors that defeated the static ACI architecture. If an adversary invents a new form of epistemic poisoning, the OCI mathematically rewrites its own validation architecture to become immune.

7 Conclusion

This paper formalizes Artificial Civilization Intelligence (ACI) as a structural necessity for planetary-scale machine intelligence. Through Project ORION, we empirically validated the framework’s Accumulated Empirical Advantage (AEA) and multi-agent coordination capabilities. However, rigorous adversarial evaluation demonstrated the fatal flaw of fixed-topology systems under epistemic attack. Consequently, we define Open-ended Civilizational Intelligence (OCI) as the requisite evolutionary leap, utilizing formal mathematical proofs to achieve safe, continuous structural plasticity.

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